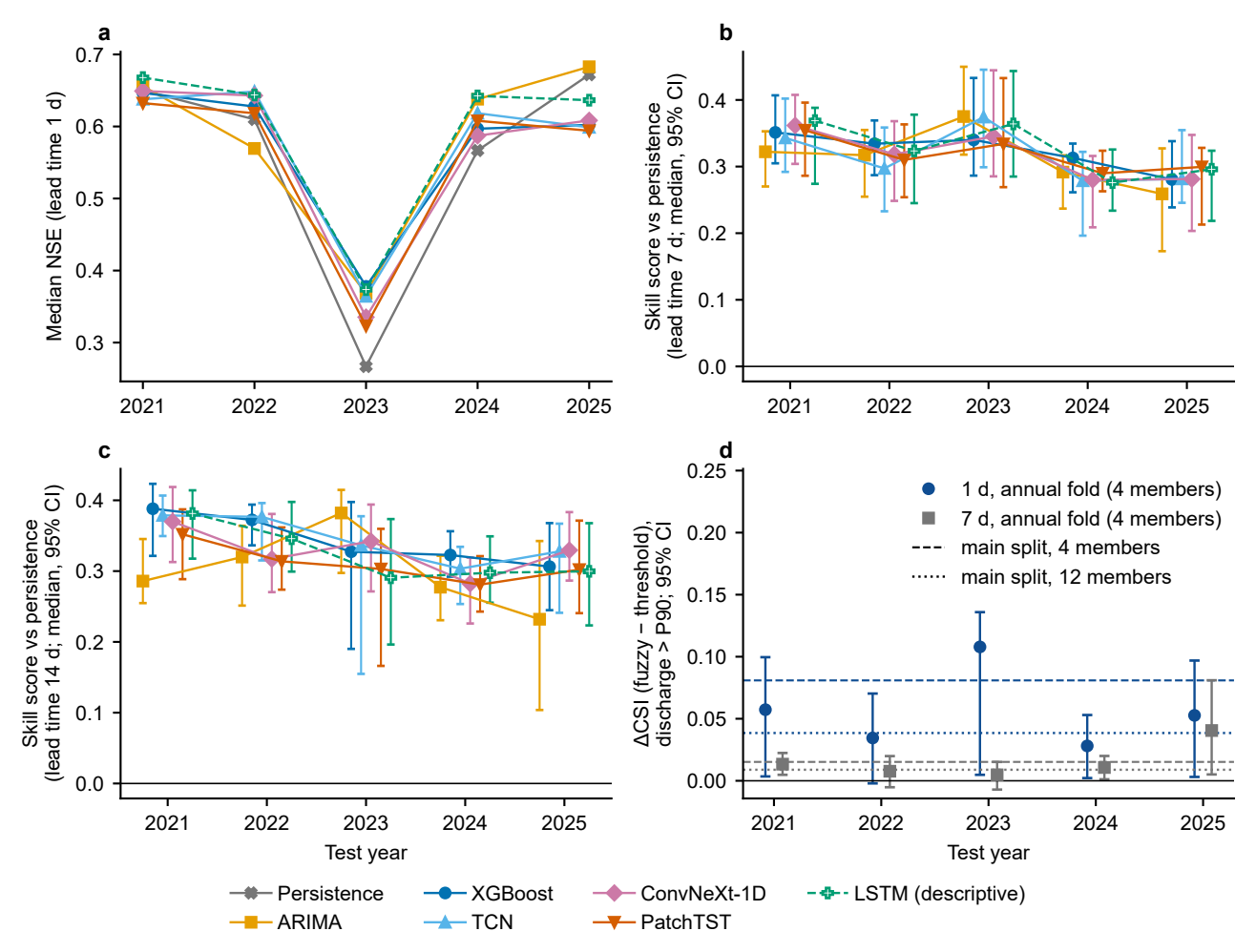

a
Median NSE (lead time 1 d)
0.7
0.6
0.5
0.4
0.3
2021
2022
2023
2024
2025
b
Skill score vs persistence
(lead time 7 d; median, 95% CI)
0.4
0.3
0.2
0.1
0.0
2021
2022
2023
2024
2025
c
Skill score vs persistence
(lead time 14 d; median, 95% CI)
0.4
0.3
0.2
0.1
0.0
2021
2022
2023
2024
2025
Test year
d
ΔCSI (fuzzy − threshold),
discharge > P90; 95% CI
0.25
0.20
0.15
0.10
0.05
0.00
1 d, annual fold (4 members)
7 d, annual fold (4 members)
main split, 4 members
main split, 12 members
2021
2022
2023
2024
2025
Test year
Persistence
ARIMA
XGBoost
TCN
ConvNeXt-1D
PatchTST
LSTM (descriptive)